IIT KHARAGPUR × upGRAD

Session 6

Testing, CI/CD & Deployment ·

Please finish this before class. It takes about 20 minutes. Session 6 is hands-on throughout — you will write tests, push code to GitHub and deploy a live URL.

Part 1 — Node.js

Check what you already have:

```
node -v
```

If it prints **v20** or higher, skip to Part 2. If it prints something older, or "command not found", install it below.

Windows

```
winget install OpenJS.NodeJS.LTS
```

Or download the LTS installer from `https://nodejs.org` and click through it.

macOS

```
brew install node
```

No Homebrew? Use the LTS installer from `https://nodejs.org`.

Linux

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt-get install -y nodejs
```

Close your terminal and open a new one, then confirm both of these:

```
node -v      # v20.x.x or higher
npm -v       # 10.x.x or higher
```

Part 2 — Git

```
git --version
```

If that fails, install it:

- **Windows:** `https://git-scm.com/download/win`
- **macOS:** `brew install git` (or `xcode-select --install`)
- **Linux:** `sudo apt install git`

Then tell git who you are. This is **required** — commits fail without it:

```
git config --global user.name "Your Name"
git config --global user.email "your@email.com"
```

Use the same email address you will use for GitHub in Part 3.

Part 3 — GitHub account and SSH key

3a. Create the account

Sign up at `https://github.com` if you do not already have an account, and verify your email address. Verification is required before you can push any code.

3b. Set up an SSH key

This is what lets you push without typing a password every time. First check whether you already have one:

```
ls ~/.ssh/id_ed25519.pub
```

If that prints a path, you already have a key — skip to 3c. If it says "No such file":

```
ssh-keygen -t ed25519 -C "your@email.com"
```

Press **Enter** three times to accept the defaults. An empty passphrase is fine for this.

Now copy the **public** key to your clipboard:

```
# macOS
pbcopy < ~/.ssh/id_ed25519.pub

# Windows (Git Bash)
clip < ~/.ssh/id_ed25519.pub

# Linux
cat ~/.ssh/id_ed25519.pub      # then select and copy the output
```

3c. Add the key to GitHub

1. Go to `https://github.com/settings/keys`
2. **New SSH key**
3. Title: anything (`laptop` is fine)
4. Paste into the Key box
5. **Add SSH key**

3d. Test it

```
ssh -T git@github.com
```

Type `yes` if it asks about authenticity. Success looks like this:

```
Hi <your-username>! You've successfully authenticated,
but GitHub does not provide shell access.
```

> **That message is correct.** "does not provide shell access" is the normal, successful response — not an error. If you see "Permission denied (publickey)" instead, the key on your machine is not the one on your account.

3e. GitHub CLI — optional, but it saves real time

```
# Windows
winget install GitHub.cli

# macOS
brew install gh
```

Then sign in:

```
gh auth login
```

Choose **GitHub.com → SSH → Login with a web browser** and follow the prompts.

This is optional. But in Session 6 it turns “create a repository and push to it” from four manual steps into **one command** — and it avoids the single most confusing error in the session, where a missing repository reports itself as an access-rights problem. Worth the two minutes.

Part 4 — Vercel account

We deploy the app to Vercel at the end of the session.

1. Go to `https://vercel.com/signup`
2. **Choose “Continue with GitHub”** — not email

> **Do not sign up with email.** An email account is not linked to your GitHub, and connecting the two afterwards mid-session is fiddly and slow. Use the GitHub button.

Authorise Vercel when GitHub asks. Nothing else to do now.

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ne last thing

Please have your AI coding agent (Claude Code, or whichever you have been using this course) installed and signed in. Several exercises ask you to prompt it directly — and the interesting part of the session is reading carefully through what it gives back.